

Track C — Autonomy and Ground Control

RescueSwarm: work package and funding request

NIDAR 2026–27 · Track 1 · Mission 1

Ask: INR 2,324 · 2 students · Coverage planner, task allocation, state machine, ground station, SITL

This track asks for the least money in the programme and carries a large share of the scored functionality. That is not an accident of accounting: the deliverables are software, the workstations are team-supplied, and the verification environment is free. It is worth stating explicitly, because a funding split that shows a near-zero line invites the assumption that the track is small. It is not — it is inexpensive.

1 Scope and ownership

This track owns **coverage planner, task allocation, state machine, ground station, sitl**. It is one of four delivery tracks defined in `docs/development-plan.md` §1.3; the integrated system argument lives in the master proposal, `docs/proposal/rescueswarm-proposal.pdf`, which this document does not restate.

2 Coverage planning over arbitrary search regions

The rulebook does not promise a rectangular search area, so the planner does not assume one. The implementation decomposes a possibly non-convex boundary into monotone cells, fills each by scanline, clips every transect to the true boundary, and routes between cells along a visibility graph shortest path rather than a straight line that might leave the region.

Verified across 13 boundary shapes including non-convex notches and slots: **0.00 m of path outside the boundary**, workload imbalance across three aircraft $\leq 0.02\%$, and 157 tests passing. Three defects were found and fixed in reaching that state — a scanline span that bridged concave notches, a missing cell decomposition, and a straight inter-cell hop that cut corners across excluded ground.

3 Autonomy, and how it is evidenced

The competition requires autonomous operation, and asserting it is not the same as showing it. The verification harness parses the flight script's own syntax tree and proves that **no MAVLink transmission occurs after the mode is set to AUTO**. That is a structural argument rather than an observational one: it does not depend on having watched a particular flight behave.

The same harness checks inter-aircraft separation, geofence containment, failsafe configuration, energy consumption against the model, and recorder clock integrity, and exits non-zero on any failure. It is re-runnable, which means the autonomy claim is regenerated rather than remembered.

A separation breach of 3.10 m was found this way and corrected to 5.34 m by staggering recovery loiter times. A 453.9 s recorder clock discontinuity was found the same way, and independently confirmed against consumed charge.

4 The ground station is deliberately unable to fly the aircraft

Requirement SYS-20 holds that the ground station must be *architecturally* incapable of originating a retask, waypoint change or arming command — not merely configured not to. This is a design property,

not a policy, and it is what makes the autonomy claim defensible under scrutiny: an operator cannot accidentally invalidate it.

5 Why the ask is small

The two workstations are team-supplied and the simulation environment is open-source. What remains is a fabricated sun hood and an observer monitor for field use. This track's real cost is student time across the full programme duration, which the funding request does not price and which the schedule must protect.

6 Funding request

Every figure below is generated from `docs/proposal/figures/track_budget.py`, which partitions the master competition budget. The four track asks plus the shared pool reconcile to the master figure exactly; that reconciliation is asserted in code and fails the build if it ever stops holding. **K** marks a line held at professional grade on purpose.

Item	INR
Sun hood + observer monitor	1,500
Subtotal	1,500
Customs duty and freight	213
GST	308
Contingency at 15%	303
TRACK C ASK	2,324

Deferred or institutionally held, and therefore not requested: Backup GCS laptop, GCS laptop. These remain capabilities the track requires; they are simply not being bought. Where a deferral carries a technical consequence it is stated in the risk table below rather than left implicit.

7 Deliverables by phase

Phase	Dates	This track delivers
P1	24 Aug – 13 Sep	SITL flying one aircraft. Planner validated against arbitrary boundaries.
P2	14 Sep – 4 Oct	Full mission in SITL. Verification harness passing.
P3	5 Oct – 18 Oct	Simulation re-baselined against the built aircraft.
P5	9 Nov – 22 Nov	Autonomous full mission, no operator input after AUTO.

8 Risks owned by this track

Risk	Sev.	Mitigation
Pad containment has zero margin	High	Recorded in the master proposal as a measured result. Recovery geometry must be re-verified whenever aircraft mass or descent rate changes.
Survey altitude is unresolved	Medium	The design point says 60 m; every flown simulation uses 40 m. Coverage timing, transect count and detection all move with it. Resolution is shared with Track D and needs a recall measurement to settle.
Simulation diverges from the built aircraft	Medium	SITL evidence is only evidence if the modelled aircraft matches the built one. Re-baseline against Track A's measured mass and thrust after P2.

9 Interfaces to other tracks

With	Dependency
Track A	Consumes measured mass, thrust and endurance. Simulation fidelity depends on receiving them after the P2 thrust-stand run.
Track B	Runs on the companion computer and commands through its MAVLink routing. Depends on the link carrying control and telemetry together.
Track D	Consumes detections and supplies the aircraft state — position, attitude, time — that each geotag is computed from.

Interfaces are where student programmes fail, because each side assumes the other owns the gap. Every row above is a two-way commitment and should be recorded as an interface control document at P1.